

Answers

VARC

1.A	2.B	3.B	4.A	5.B	6.C	7.A	8.C
9.C	10.B	11.D	12.2	13.A	14.B	15.D	16.D
17.A	18.B	19.C	20.1	21.B	22.B	23.B	24.C

LRDI

25.C	26.4	27.C	28.A	29.45	30.A	31.2	32.B
33.8	34.4	35.A	36.3	37.5	38.D	39.D	40.A
41.B	42.B	43.3	44.2	45.4	46.D		

Quant

47.D	48.A	49.38	50.A	51.6	52.B	53.C	54.217
55.B	56.31	57.10	58.A	59.31	60.B	61.B	62.66
63.D	64.15	65.D	66.C	67.A	68.C		

Explanations

VARC

1. A

Option A is the correct answer.

The passage mentions that medieval craft guilds were monopolies that resisted new entrants and required long apprenticeship periods, which could stifle innovative spirit. Similarly, the ethos of mass production prioritizes efficiency and consistency, which often comes at the expense of creativity. Therefore, Option A could be inferred.

Option B: While it is true that medieval guilds restricted entry through strict rules, the passage does not suggest that mass production involves such restrictions.

Option C: The passage does not discuss whether mass production or medieval guilds employed egalitarian processes.

Option D: The passage does not mention that medieval guilds or mass production focused excessively on product quality.

[▶ VIDEO SOLUTION](#)

2. B

Option B is the correct answer.

The passage mentions that medieval craft guilds were monopolistic and hierarchical, resisting new entrants and imposing long apprenticeships, which could stifle innovation. It also warns that modern craft workers “can thrive... only if they don't get too organised,” supporting option B.

Option A: This is an extreme interpretation. The passage does not argue that supporting crafts is the only way to retain the creativity intrinsic to their production.

Option C: The Arts and Crafts movement is described as a reaction against the “American system” and the rise of mass production, not an inspiration drawn from it.

Option D: Agile movement is praised for prioritizing creativity and collaboration, whereas medieval craft guilds are described as hierarchical and restrictive, which stifled innovation. These are contrasting ideas. Therefore, agile movement cannot be a throwback to the tenets (principles) of medieval craft guilds)

 VIDEO SOLUTION

3. B

Option B is the correct answer.

The author says “*In a world where products and services often have to pass through regulatory hoops, large companies will usually have the advantage.*” From this, we can infer that the author doubts whether crafts can create substantial employment opportunities because smaller craft businesses may struggle to compete with larger companies due to regulatory barriers.

Option A: The passage does not focus on the low scale of crafts production as the primary obstacle. Instead, the author emphasizes regulatory challenges and not the scale.

Option C: The passage does not argue that workers wouldn't want to pursue crafts-related work, just that retraining them for these roles might be difficult.

Option D: The passage mentions that craft guilds resisted new entrants, but it does not suggest that they are unlikely to accept large number of trainees. The passage suggests that craft workers can thrive in the modern era, but the challenge lies in how modern crafts are organized and their potential to scale up in a competitive market.

 VIDEO SOLUTION

4. A

Option A is the correct answer.

While the passage mentions the premium for high-quality crafted goods and the contrast between mass production and craftsmanship, it does not mention that there is support for “individual creations” as opposed to mass-produced goods. The focus is on quality and sustainability rather than explicitly advocating for individuality in products.

Option B: This is mentioned. The passage talks about consumers wanting to reduce environmental impact by supporting local workers or recycling goods, which reflects growing concerns over mass production.

Option C: This is mentioned. The passage describes a market of consumers who pay a premium for high-quality, hand-crafted goods.

Option D: This is mentioned. The passage highlights a market where consumers buy from local workers to support the community and reduce environmental impact.

 VIDEO SOLUTION

5. B

Option B is the correct answer.

The paragraph states that animals share many emotions with humans, such as joy, happiness, empathy, and grief, because of shared brain structures, particularly in the limbic system, which is responsible for emotions in both humans and animals. This is the key point that ties together animals' intelligence and emotional capacity, as discussed in the passage.

Option A: The passage does not attribute emotions to sensory and motor abilities. The emphasis is on brain structures, not sensory abilities.

Option C: While the passage states that animals share emotions with humans, this option fails to capture the reason behind this, i.e. shared brain structures. It misses the point that makes the emotional similarity possible, which is central to the passage's message.

Option D: The passage discusses animals' sensory abilities but does not suggest that their intelligence is superior to humans'.

 VIDEO SOLUTION



6. C

Option C is the correct answer.

The passage suggests that the idea of 'homo economicus' assumes that people have “preferences” that determine their choices. Deciding whether to spend limited money on food or warmth illustrates this rational decision-making based on individual preferences. This aligns with option C

Option A: The passage does not suggest that 'homo economicus' is not influenced by others' preferences. It only states that this model assumes people have their own preferences that determine their actions.

Option B: This is wrong as the passage states that economists like Gary Becker, associated with the homo economicus model, did not borrow or collaborate with other disciplines.

Option D: The passage mentions applying economic reasoning to nonmarket domains but does not tie this specifically to homo economicus.

7. A

Option A is the correct answer.

The tone of the phrase “almighty discipline” suggests mild sarcasm regarding economics’ earlier dominance and self-containment. The statement reflects a shift in economics from being insular and self-assured to being more open to interdisciplinary borrowing. The author describes earlier economists, like Gary Becker, as imperialistic and unwilling to incorporate ideas from other disciplines. In contrast, contemporary economists like Thomas Piketty are portrayed as reaching out and collaborating.

Option B: The passage mentions that economists are now collaborating with other disciplines, but it does not suggest this is due to their inability to predict market behaviour. The focus is on how the discipline has shifted over time, not any predictive failures.

Option C: The author does not criticize economic tools for managing crises or say that economics as a discipline has fallen. The passage highlights a transformation toward openness, not a downfall.

Option D: The author is not critical of economists collaborating with other disciplines. Instead, the passage presents this shift as a positive change, contrasting it with the earlier insularity of the discipline.

8. C

Option C is the correct answer.

The author critiques Schiller for focusing on the direct relationship between emotions, perceptions, and behaviour while excluding the mediating role of institutions such as political parties, media organizations, and lobby groups. The passage states that these institutions historically played a vital role in framing perceptions and legitimizing interests, which Schiller’s narrative leaves out.

Option A: The passage suggests that media and politics are key intermediaries in shaping perceptions and behaviour and not a marginal role.

Option B: The passage does not suggest that Schiller actively denigrates or diminishes the role of institutions. The issue is his omission of institutions, not his critique of their importance.

Option D: The passage mentions that storytelling is part of Schiller’s approach but does not critique him for relying excessively on it. The critique is about the exclusion of institutions, not the use of storytelling.

9. C

Option C is the correct answer.

The first paragraph describes how Becker projected economics into non-market domains like crime and domesticity, analyzing them with economic tools. It also states that he did not borrow ideas or perspectives from fields such as anthropology or history. Option C aligns with this.

Option A: The passage mentions that Becker applied economic principles to non-market phenomena like crime and domesticity, but it does not state or imply that he benefitted from this.

Option B: The passage explicitly contradicts this idea, stating: “*At the same time, he did not let other ways of thinking enter his own economic realm: for example, he did not borrow from anthropology or history.*” Becker did not borrow from other disciplines, so this option is incorrect.

Option D: The passage does not state that Becker actively guarded economics against outside influence. Instead, it focuses on Becker’s one-sided application of economics to other areas, without accepting external perspectives.

10. B

The sentence would best fit in B) Option 3.

Option 3 is the most logical position because it continues the description of Renaissance music’s characteristics, which was mentioned in the sentence before blank 3. The missing sentence fits perfectly here, as it discusses how Renaissance music handles emotions in a balanced and moderate manner, aligning with the flow and tone of the preceding sentences.

The sentences around Option 1 and Option 2 discuss the historical context of English madrigals and their cultural significance, so adding the sentence there would disrupt the historical narrative.

Option 4 may appear plausible, but the sentence discussing the balanced and moderate nature of music after asserting that extremes do not occur would be redundant. The given sentence should precede the one addressing extremes to maintain logical flow.

CAT Percentile Predictor



11. D

We can observe that the only spot where the given sentence would fit is Blank 4. This sentence in question emphasises the broader, global significance of Central Asia’s role; it reflects the idea that understanding the Silk Roads provides clarity on historical and global interconnectedness.

Since the given sentence shifts focus from describing the historical and cultural importance of the Silk Roads to their global implications, which is broader than the current discussion, placing it in any of Blanks 1, 2, or 3 interrupts the flow between the statements outlining certain features linked to the Silk Roads. For example, Blank 1 is a poor fit since the succeeding sentence mentions “These countries,” which refers to the “developing countries” mentioned in the sentence preceding the blank. Similarly, “They” in the sentence after Blank 2 refers to routes/Silk Roads described earlier. We can eliminate Blank 3 using a similar logic;

the sentence after this blank establishes the metaphor of the Silk Roads as a central nervous system, emphasizing their global connectivity. The given sentence doesn't align with the focus on the Silk Roads' tangible impacts or the metaphorical depiction of their connectivity; therefore, placing it in Blank 3 would shift the focus prematurely to understanding their role globally.

[VIDEO SOLUTION](#)

12.2

Sentence 2 is the odd one out.

The sequence 5-3-4-1 forms a coherent paragraph.

Sentence 5 introduces the issue of the puzzling decline in public transport despite growing urban populations and employment. Sentence 3, then explains some of the reasons for the decline, like worsening services, terrorist attacks, and rising fares. Sentence 4 adds a structural explanation, stating that public transport is being squeezed as people travel less due to technology. Sentence 1 provides more details about the alternatives to public transport, which are contributing to the decline in usage.

Sentence 2 talks about how more people using buses or trains improves the service, but this idea is not directly relevant to the overall discussion in the passage, which focuses on the decline of public transport usage. It shifts the focus toward an increase in usage improving services, which contradicts the central theme of decline.

[VIDEO SOLUTION](#)

13.A

Option A is the correct answer.

The passage mentions how bandicoots contribute to the ecosystem by digging, which traps moisture and aids in seed germination. This activity helps restore the desert ecosystem damaged by cattle. Their role as "ecosystem engineers" stems from these positive environmental impacts.

Option B: Although efforts to preserve the bandicoot species are ongoing, the passage does not link their new nickname to these efforts. The name "ecosystem engineers" specifically reflects their environmental contributions rather than conservation measures.

Option C: While the passage mentions a population increase due to rainfall, the new nickname is unrelated to this growth. Instead, it is tied to their environmental engineering role.

Option D: This is a wrong interpretation because there is no mention in the passage of the bandicoots affecting rainfall.

[VIDEO SOLUTION](#)

14.B

Option B is the correct answer.

The passage does not mention the bandicoots' use of camouflage as a survival technique. While their shelters may be camouflaged (hidden), the bandicoots themselves are not described as using camouflage directly.

Option A: This is correct. The passage mentions their long, slender snouts and their digging behaviour, which allows them to create shallow shelters in the desert.

Option C: This is correct. The nickname "zebra rat" comes from their appearance, and their slender snout and backward-facing pouch for carrying joeys are described in the passage.

Option D: This is also correct. The passage mentions these features as characteristics of the western barred bandicoot.

[VIDEO SOLUTION](#)

15.D

Option D is the correct answer.

The 'enclosures' are mentioned as fenced areas cleared of invasive rabbits and feral cats. The term "enclosures" points to the intentional exclusion of these invasive species to create a safe environment for the bandicoots and other native animals.

Option A: While the bandicoots help in restoring the cattle damaged landscape, the term 'enclosure' does not relate to it.

Option B: The enclosures are cleared of feral cats, but the passage does not mention removing large bilbies, which are actually part of the controlled environment. The enclosures themselves are specifically to protect the bandicoots from cats and rabbits, not bilbies.

Option C: The enclosures are not about making an area entirely predator-free for all species. Instead, the purpose is to create controlled environments where invasive species like rabbits and feral cats are removed. Predators are still present in the Wild Training Zone, where bandicoots and other marsupials learn to coexist and evade predators.

[VIDEO SOLUTION](#)



16.D

Option D is the correct answer.

Option D captures the main idea of the passage. It reflects the near-extinction of the western barred bandicoot due to invasive species and highlights the conservation efforts using survivors from Shark Bay islands.

Option A: This is not entirely true. The western barred bandicoot did not go extinct; instead, it survived in small numbers on two predator-free islands. This option incorrectly asserts total extinction and ignores the ongoing efforts to revive the species.

Option B: This is a distortion. While the colonists' negligence and the nicknames they gave reflect their disregard, the passage clearly attributes the near-extinction of bandicoots to ecological disruptions caused by invasive species, not merely the colonists' attitudes.

Option C: This generalizes the issue and does not focus specifically on the western barred bandicoot, the subject of the passage. Furthermore, it does not highlight the rescue efforts which are central to the passage.

VIDEO SOLUTION

17. A

Option A is the correct answer.

The passage emphasizes that cartographers now should pay attention to the usability of maps due to the evolving expectations of map readers. The key point is that technological developments have made users more demanding, leading cartographers to focus on how efficient, effective, and appreciated their maps are.

Option B: While it is true that cartographers are focused on usability, the passage does not mention specific experiments or evaluation methods.

Option C: This option suggests that maps are being used for a variety of reasons, which is not mentioned in the passage. The focus of the passage is on the demanding nature of modern map readers and not on the reasons for which maps are used.

Option D: While new technological developments are mentioned, the passage does not state that cartographers are experimenting with these innovations in their maps.

VIDEO SOLUTION

18. B

Let's evaluate each option:

Option 1: The given sentence works well as an introduction. It sets the stage by comparing the brain's organization to familiar spaces like a home office or medicine cabinet. This analogy smoothly transitions into the next sentence about the brain's "haphazard and disjointed" architecture, making it a natural fit here.

Option 2: Inserting the sentence here would disrupt the flow. This part focuses on the brain's evolved systems and how evolution works. The analogy to home offices feels out of place and does not connect coherently with the subsequent sentences.

Option 3: This gap already includes a metaphor—the brain as "a big, old house with piecemeal renovations." Adding the sentence here would be redundant, as it repeats the idea of disorganization without adding new insight.

Option 4: This section discusses how evolution doesn't design harmonious systems. The sentence about the brain's organization doesn't align well with this abstract explanation, making it feel disjointed and out of context.

Option 1 is the best fit as it introduces the idea clearly, aligns with the paragraph's theme, and transitions seamlessly into the discussion of the brain's complexity.

VIDEO SOLUTION

19. C

Option C is the correct answer.

This option captures the main idea of the passage that some codes, like language and visual signs, are so commonly used that they appear natural and conceal the process of how they were created.

Option A: The passage does not suggest that early learning is why codes appear natural. The cause-and-effect relationship is incorrectly stated here.

Option B: This option misinterprets two key aspects of the passage. First, the idea that certain codes are "made to appear universal" is somewhat misleading because the passage doesn't claim that codes are deliberately made universal; instead, it describes how codes, through habituation and widespread use, come to feel "natural". Second, the phrase "Ideology aims to hide the mechanism of coding" is not supported by the passage. The passage suggests that the naturalization of codes leads to the illusion of transparency and naturalness, which conceals the mechanisms of coding, but it doesn't explicitly discuss ideology as a force that intentionally hides these mechanisms.

Option D: This option is incorrect because the passage doesn't claim that all codes have a natural origin. It states that codes become naturalized through use, not that they were naturally originating from the start.

VIDEO SOLUTION

20. 1

Sentence 1 is the odd sentence.

The correct sequence is 4-2-3-5.

Sentence 4 introduces the historical significance of Peter Singer's *Animal Liberation*, which set the stage for the discussion of animal rights in contemporary philosophy.

Sentence 2 expands on Singer's philosophy, explaining his utilitarian view and how it contrasts with other indirect moral views on animal rights.

Sentence 3 then explains the core of Singer's argument, that animals, like humans, have significant interests and deserve moral consideration.

Sentence 5 concludes Singer's argument by stating that humans should treat animal interests equally to human interests, leading to moral duties towards animals.

Sentence 1 is the odd one out because it does not focus on Peter Singer's views on animal rights and the moral obligations humans have towards animals, which is the main theme of the paragraph.

VIDEO SOLUTION

21. **B**

Option B is the correct answer.

The passage argues that, despite the advances in digital distribution and storage of films (via streaming services or digital purchases), access to art is becoming more fragile. It mentions how content can disappear from platforms, how digital files deteriorate over time, and how rights agreements can limit access based on geographical location. The idea that technology and platform control lead to difficulties in maintaining access captured in Option B

Option A: The passage does not advocate changing the understanding of art as immutable or easily available. Hence, this is wrong.

Option C: This is an overstatement of the passage's argument. While the passage briefly touches on the idea of retroactive changes to works like Stranger Things, it does not present these changes as inherently "dangerous."

Option D: The passage does not argue for the availability of art in the cultural commons in perpetuity. Instead, it highlights how access is controlled by platforms and technological challenges rather than making a broader ideological statement about cultural commons.

[VIDEO SOLUTION](#)

22. **B**

Option B is the correct answer.

Option B would invalidate the main argument because it directly addresses the issue raised in the passage, i.e., the lack of permanent access to digital content. The passage highlights concerns about the temporary and restricted nature of digital ownership. If studios and streaming services committed to providing perpetual and platform-independent access, it would resolve the problem of content being removed or restricted, making the author's argument about the instability of digital media irrelevant.

Option A: This would not invalidate the argument because the passage mentions that Blu-ray discs have a theoretical shelf life but acknowledges that their durability depends on storage conditions and the availability of playback equipment.

Option C: This option doesn't directly invalidate the argument either. While VPNs might help users bypass geo-restrictions, it doesn't address the broader issue of digital ownership and the fragility of digital rights, especially the fear of losing access to content due to different rights agreements. The passage focuses on the unreliability and restrictions of digital ownership, not just geographical access.

Option D: While this option touches on the potential for preserving digital content, it doesn't directly address the problem raised in the passage: the lack of permanent, independent access to content.

[VIDEO SOLUTION](#)

23. **B**

Option B is the correct answer.

The passage contrasts the past and present by mentioning that, in the past, films were considered "as good as gone" once they left the cinema, meaning they were ephemeral and not readily accessible afterwards. Whereas, today's audience expects ongoing access to films well beyond their initial cinema run, thanks to technological advancements like streaming services and digital media. This shift in expectations is what the passage implies when referencing the previous era's ephemerality versus today's more lasting availability.

Option A: This is not the main point. While it may be true that people accepted films as temporary, the passage emphasizes today's expectations rather than discussing past acceptance.

Option C: The passage does not mention technology improvements. It focuses more on audience expectations or belief that films should now be available beyond just the cinema.

Option D: While the passage suggests that audiences may expect films to remain accessible, it does not claim there is no reason for studios to remove access. The passage acknowledges that financial motives may lead to films being removed from platforms.

[VIDEO SOLUTION](#)

24. **C**

Option C is the correct answer.

The passage highlights that the practice of streaming services, like Netflix, editing old episodes of Stranger Things retroactively raised concerns. Altering a popular show's content without consent or transparency supports the concern that such platforms can tamper with or erase parts of culture at their discretion rather than preserve them as they were originally created.

Option A: This option doesn't fully address the specific concern raised by the Stranger Things editing example. The example focuses on altering the content, not controlling access.

Option B: The example concerns the possibility and practice of editing content on streaming platforms, not whether unsubstantiated reports cause distrust.

Option D: The example of Stranger Things is not necessarily about changing films to suit new tastes or technology. Rather, it's about the platform's ability to alter existing content without transparency or input from the original creators.

[VIDEO SOLUTION](#)

LRDI

25. **C**

Writing down the values given in the candlestick chart in the form of a table for ease of calculation,

Stock	Opening	Close	High	Low
A	2200	1800	2400	1200
B	2000	1700	2000	1400
C	800	1200	1400	800
D	500	1000	1200	300
E	1300	1100	1400	1100
F	1800	1600	2000	1200
G	1200	1700	1900	1000

We are given that, Daily Share Price Variability (SPV) is defined as $(\text{Day's high price} - \text{Day's low price}) / (\text{Average of the opening and closing prices during the day})$

Calculating it for the four options,

Stock F: $800/1700=8/17$

Stock A: $1200/2000=3/5$

Stock D: $900/750=90/75=6/5$

Stock C: $600/1000=3/5$

Clearly Stock D has the highest SPV.

[VIDEO SOLUTION](#)



26.4

Writing down the values given in the candlestick chart in the form of a table for ease of calculation,

Stock	Opening	Close	High	Low
A	2200	1800	2400	1200
B	2000	1700	2000	1400
C	800	1200	1400	800
D	500	1000	1200	300
E	1300	1100	1400	1100
F	1800	1600	2000	1200
G	1200	1700	1900	1000

We are given that, Daily Share Price Variability (SPV) is defined as $(\text{Day's high price} - \text{Day's low price}) / (\text{Average of the opening and closing prices during the day})$

Calculating it for the stocks

Stock A: $1200/2000=3/5$

Stock B: $600/1850=60/185$

Stock C: $600/1000=3/5$

Stock D: $900/750=90/75=6/5$

Stock E: $300/1200=1/4$

Stock F: $800/1700=8/17$

Stock G: $900/1450=90/145$

We need to check for stocks greater than 0.5 on that day,

Stock A, Stock C, Stock D, Stock G have SPV greater than 0.5 that day.

Hence, the answer is 4.

[VIDEO SOLUTION](#)

27. C

Writing down the values given in the candlestick chart in the form of a table for ease of calculation,

Stock	Opening	Close	High	Low
A	2200	1800	2400	1200
B	2000	1700	2000	1400
C	800	1200	1400	800
D	500	1000	1200	300
E	1300	1100	1400	1100
F	1800	1600	2000	1200
G	1200	1700	1900	1000

Daily loss for a share is defined as (Opening price - Closing price) / (Opening price)

Calculating this for the options:

Stock A: $400/2200=2/11$

Stock B: $300/2000=3/20$

Stock F: $200/1800=1/9$

Stock G gained money that day

Hence Stock A has the highest Daily Loss.

[VIDEO SOLUTION](#)

28. A

Writing down the values given in the candlestick chart in the form of a table for ease of calculation,

Stock	Opening	Close	High	Low
A	2200	1800	2400	1200
B	2000	1700	2000	1400
C	800	1200	1400	800
D	500	1000	1200	300
E	1300	1100	1400	1100
F	1800	1600	2000	1200
G	1200	1700	1900	1000

There are three bullish shares, C D and G

Lets say a trader buys one share of each of these stocks, and sells them at their day's high

One share of C at opening is 800, sells at 1400

One share of D at opening is 500, 1200

One share of G at opening is 1200, 1900

Total Investment is 2500, and total money after selling is 4500

That is an 80% return since, $\frac{(4500 - 2500)}{2500} = 0.8$

[VIDEO SOLUTION](#)

Explanation [29 - 32]:

We can note down the data from the chart into the form of a table, that gives us,

Surfer/Blogger	A	B	C	D	Total
M	10	0			30
N	25	0			30
O	0	0			30
P	5	25			30
X	0	0			30
Y	5	20			30
Total	45	45	45	45	180

We are told that D receives more stars than C from Y. Considering Y has already given 25 stars, it will give 0 stars to C and 5 stars to D.

The only two surfers who have not given any stars to A or B is O and X, and these are the two surfers to give all of their stars to a single blogger.

We are also told that M gives different stars to the four bloggers,

Since he has already given 0 and 10, the remaining distinct stars should add up to 20. The only numbers that are remaining that add up to 20 are 5 and 15.

We know that X rewards one of C or D 30 stars and O rewards one of C or D 30 stars. Given that, M could not have rewarded D 15 stars, since Y rewarded D 5 stars, and D is also going to be rewarded 30 stars by O or X, and since the total is same for all, which is 45. This is not possible.

This means that, M rewarded C 15 stars and D 5 stars.

This gives us two cases,

Case-1

Surfer/Blogger	A	B	C	D	Total
M	10	0	15	5	30
N	25	0	0	5	30
O	0	0	30	0	30
P	5	25	0	0	30
X	0	0	0	30	30
Y	5	20	0	5	30
Total	45	45	45	45	180

Case-2

Surfer/Blogger	A	B	C	D	Total
M	10	0	15	5	30
N	25	0	0	5	30
O	0	0	0	30	30
P	5	25	0	0	30
X	0	0	30	0	30
Y	5	20	0	5	30
Total	45	45	45	45	180

We can use these two cases to answer the questions,

29.45

D was rewarded 45 stars in total.

[VIDEO SOLUTION](#)

30.A

D received 5 stars from Y.

[VIDEO SOLUTION](#)



CAT Daily Targets

Step by step, heading towards 99+ percentile



31.2

M distributed among 3 bloggers, N among 2 bloggers, O among 1, P among 2, X among 1, Y among 3

Hence 2 surfers distribute their stars among 2 bloggers.

[VIDEO SOLUTION](#)

32.B

There are two cases formed since we cannot identify for certain two whom did O reward the stars to,

We can identify the number of stars rewarded by M to C.

Hence only statement 1 can be identified uniquely.

[VIDEO SOLUTION](#)

Explanation [33 - 37]:

We are told that there are six teams, that are divided into two groups.

Teams in the same group will play each other only once, and teams in different group will play each other twice.

Calculating the combinations, there is going to be 3C_2 games among teams in the same group among them, and since there is two groups, total such games will be 6.

Now, teams in different group play each other twice. Calculating the combinations for this,

From the first group, a team can be chosen in three ways, and from the second group, a team can be chosen in three ways. Total ways two teams from different groups can play each other is 3×3 which is 9. And since they play each other twice, that is 9×2 games of this combination.

Total number of games is $18 + 6 = 24$

It is given that each team plays one game in each round, that means there is going to be 3 matchups in each round. And given, there is 24 games to played in this format, the number of rounds will be $24/3 = 8$

The tournament will have 8 rounds.

Now that we know there are going to be 8 rounds in the tournament, let us identify the teams in a particular group, that will help us build the matchups.

We are told that Round 8 teams from different groups play each other, and Teams 1 and 5 play only once. This means, 1 and 5 have to be on the same group. It is also told that 4 and 6 play each other twice, that means 4 and 6 have to be in different groups. Looking at the matches from Round 8 that is given to us, 3 played 6 and 2 played 5. We already know 1 and 5 are in the same group, so 2 must be in the other group. Among 3 and 6, if we were to place 3 in the group with 1 and 5, 4 and 6 would have to be in the same group, which is not possible, hence 6 is with 1 and 5, giving us the final combination of groups.

Group-1	Group-2
1	2
5	3
6	4

Now, using the given information to build the matchups for the 8 rounds.

Round-1	Round-2	Round-3	Round-4
4 vs 6	4 vs 6	3 vs 4	
	1 vs 5		
Round-5	Round-6	Round-7	Round-8
	1 vs 6		3 vs 6
			2 vs 5

Rounds marked with the same colour represent the fact that the matchups are identical. Now we know that each team plays a game in each round, we know 2 out of the 3 matches for Round 2 and 8, and we can identify the third matchups as well. Giving us this resulting table.

Round-1	Round-2	Round-3	Round-4
4 vs 6	4 vs 6	3 vs 4	
	1 vs 5		
	2 vs 3		
Round-5	Round-6	Round-7	Round-8
3 vs 6	1 vs 6		3 vs 6
2 vs 5			2 vs 5
1 vs 4			1 vs 4

We are told that Round 4 and 7 are identical, that means they are the matchups between teams from two different groups, We look at the matchups that are remaining among the 6 teams where both the games are left to play.

Right away we identify that 6 is yet to play 2 twice and 5 once. We are looking for teams playing twice, so both Round 4 and 7 has a matchups between 2 and 6. This means 6 will play 5 in Round 3 and using that we can identify the third matchup in Round 3 as well.

Round-1	Round-2	Round-3	Round-4
4 vs 6	4 vs 6	3 vs 4	2 vs 6
	1 vs 5	5 vs 6	
	2 vs 3	1 vs 2	
Round-5	Round-6	Round-7	Round-8
3 vs 6	1 vs 6	2 vs 6	3 vs 6
2 vs 5			2 vs 5
1 vs 4			1 vs 4

Now, we can identify that 1 is yet to play 3 both the times, 2 once. And we are looking for teams playing each other twice, so we can fill in the remaining values in the table as well.

Round-1	Round-2	Round-3	Round-4
4 vs 6	4 vs 6	3 vs 4	2 vs 6
1 vs 2	1 vs 5	5 vs 6	1 vs 3
3 vs 5	2 vs 3	1 vs 2	4 vs 5
Round-5	Round-6	Round-7	Round-8
3 vs 6	1 vs 6	2 vs 6	3 vs 6
2 vs 5	2 vs 4	1 vs 3	2 vs 5
1 vs 4	3 vs 5	4 vs 5	1 vs 4

. This is the final table.

33.8

The tournament will have 8 rounds.

[VIDEO SOLUTION](#)

34.4

The team that played Team 1 in Round 5 is 4.

[VIDEO SOLUTION](#)

35.A

2, 3 and 4 were part of the same group. 5 is the answer.

[VIDEO SOLUTION](#)



36.3

The team that played Team 1 in Round 7 is 3.

[VIDEO SOLUTION](#)

37.5

Team that played Team 6 in Round 3 was Team 5.

[VIDEO SOLUTION](#)

38.D

If both Ramya and Amiya run staid campaigns, the intensity of each staid campaign is 1
So total number of voters that would vote for them if they focused on issues will be $20 \times (1+1)\% = 40\%$

This means if they had both ran regarding issues, then they would get 20% of votes each.

We are told that both of them run attacking campaigns,

And the rule for mutual attacking campaign is 10% of voters who would have voted for each candidate will not vote.

That means 10% of 20% of each candidate will not vote, now that it is a mutually attacking campaign.

That means, each candidate receives 18% of the votes.

Total votes received is 36%.

[VIDEO SOLUTION](#)

39.D

We want the minimum vote share, that means both the candidates run staid campaigns.

And, both the campaigns should be attacking, since we see that if one candidate runs an attacking campaign whereas the other candidate runs an issues campaign, a fair number of voters get transferred to the other candidate's vote share.

This points to us to a scenario where both the campaigns are staid and attacking, which is nothing but the scenario described in the previous question.

If both Ramya and Amiya run staid campaigns, the intensity of each staid campaign is 1
So total number of voters that would vote for them if they focused on issues will be $20 \times (1+1)\% = 40\%$

This means if they had both ran regarding issues, then they would get 20% of votes each.

We are told that both of them run attacking campaigns,

And the rule for mutual attacking campaign is 10% of voters who would have voted for each candidate will not vote.

That means 10% of 20% of each candidate will not vote, now that it is a mutually attacking campaign.

That means, each candidate receives 18% of the votes.

Total votes received is 36%, which is the minimum possible.

[VIDEO SOLUTION](#)

40.A

Amiya runs a campaign on issues, and we need to find the maximum vote share that she can get.

We are trying to maximise the number of voters for Amiya, that means Amiya needs to run a vigorous campaign.

Since, we are trying to increase the vote share for Amiya, we want as many voters as possible transferred from Ramya's share to Amiya's votes.

As the number of votes a candidate receives is proportional to the intensity level of the campaign, we want Ramya also to run a vigorous AND attacking campaign, so that her votes are transferred to Amiya.

In this scenario we have $20 \times (2+2)\%$ of the people voting, 80% of the people. Of that, if both had ran issues campaign, they would have each received 40% of the votes.

Now, we want Ramya to run an attacking campaign, where 20% of the people that would have voted for her vote for Amiya.

So 20% of 40% of the votes are transferred to Amiya. That is 8% of the votes. And we are also told that, 5% that would have voted for her dont vote, so 5% of 40% dont vote, that is 2% of the voters.

Final Tally is Amiya gets 48% of the votes, and Ramya gets 30% of the votes.

[VIDEO SOLUTION](#)

41. B

We are looking for the minimum possible number of votes that Ramya can get when she runs an attacking campaign.

To minimise the number of votes, we can have Ramya run a staid campaign to minimise the votes, so minimum intensity, which will get her 20% of the votes if she ran with issues. Now that she is running with attacking, she will loose 20% of the votes to Amiya and 5% of the votes will not vote anymore.

That is a total 25% loss. Remaining votes she will get is 75% of the 20% which will leave her with 15% of the votes.

[VIDEO SOLUTION](#)

42. B

We are looking for the minimum possible number of votes that Ramya can get and maximise the number of votes that Amiya can get.

We can borrow the scenario from the previous question where Ramya runs an attacking campaign, and we minimised the number of votes she can get.

To minimise the number of votes, we can have Ramya run a staid campaign to minimise the votes, so minimum intensity, which will get her 20% of the votes if she ran with issues. Now that she is running with attacking, she will loose 20% of the votes to Amiya and 5% of the votes will not vote anymore.

That is a total 25% loss. Remaining votes she will get is 75% of the 20% which will leave her with 15% of the votes.

And to maximise the number of votes Amiya can get, we will have her run an vigorous issues campaign, which will give her 2x20% of the votes, that is 40% of the votes. And since Ramya has been running an attacking campaign, 20% of her votes are transferred to Amiya. 20% of the 20% of the votes which is 4% that were going to Ramya will now go to Amiya. That will bring up Amiya's tally up to 44% leaving Ramya's tally at 15%.

The difference in the votes will be 44-15=29%.

This is the maximum possible vote difference between the two candidates that is possible.

[VIDEO SOLUTION](#)

Explanation [43 - 46]:

Writing down the data with us in a table form.

	Asia	Europe	Rest of World	Total
Dheeraj	3	7	1	11
Samantha	0	9	4	13
Nitesh	1	6	12	19
Total	4	22	17	43

But this includes countries which have been visited by one person, two people and three people. Essentially, there is an overlap of countries.

It is given to us that 32 distinct countries were visited among the three people,

Using that we can write down the equations,

$$I + II + III = 32 \dots (1)$$

$$I + 2II + 3III = 43 \dots (2)$$

Where I, II, and III denote the number of countries visited exactly by 1 of them, 2 of them and all of them.

We are told that USA(ROW) is the only country which was visited by all three people, that means $III = 1$

Subtracting Equation 1 from 2 we get, $II + 2III = 11$

And since $III=1$ we get $II = 9$

So, we are looking for 9 countries visited by exactly 2 people, and 22 countries visited by exactly 1 person.

Using the information from the problem set, we can note down the following,

Combination	Asia	EU	ROW	Total
Dheeraj				
Samantha				
Nitesh				
Dheeraj and Nitesh	1(China)			
Dheeraj and Samantha		1(France)		
Samantha and Nitesh				
All Three	0	0	1(USA)	
Total				32

We can see that Dheeraj visited only 1 country in ROW so every other value in that column has to be zero for wherever Dinesh is present. And Samantha has not visited any country in Asia so every value with Samantha present in Asia should be zero as well.

Combination	Asia	EU	ROW	Total
Dheeraj			0	
Samantha	0			
Nitesh				
Dheeraj and Nitesh	1(China)	0	0	1
Dheeraj and Samantha	0	1(France)	0	1
Samantha and Nitesh	0			
All Three	0	0	1(USA)	
Total				32

We have the following distribution, we are looking for 9 countries visited by exactly 2 people. And we have identified 2 countries, that means, 7 were countries were visited by both Samantha and Nitesh only. And given that half the countries visited by both of them are in Europe, and total number of countries visited by both of them is $7(\text{only two of them}) + 1(\text{USA}) = 8$, they visited 4 countries in Europe together and 3 countries in ROW together. And using that information, we can fill in the rest of the table as well. Since we know the number of countries visited by each of them in a particular continent.

Combination	Asia	EU	ROW	Total
Dheeraj	2	6	0	8
Samantha	0	4	0	4
Nitesh	0	2	8	10
Dheeraj and Nitesh	1(China)	0	0	1
Dheeraj and Samantha	0	1(France)	0	1
Samantha and Nitesh	0	4	3	7
All Three	0	0	1(USA)	1
Total	3	17	12	32

43.3

How many countries in Asia were visited by at least one of Dheeraj, Samantha and Nitesh is 3.

[VIDEO SOLUTION](#)

44.2

Number of countries visited by only Nitesh is 2.

[VIDEO SOLUTION](#)

45.4

Number of countries in ROW visited by both Nitesh and Samantha is $3+1=4$.

[VIDEO SOLUTION](#)



46.D

Number of countries in EU visited by exactly 1 person is 12.

[VIDEO SOLUTION](#)

Quant

47.D

Set $A = \{2, 3, 5, 7, 11, 13\}$ so $|A| = 6$

Set $B = \{1, 8, 27\}$ so $|B| = 3$

Without any restrictions, each element in A can map to any of the 3 elements in B. Thus, the total number of functions is: $3^6 = 729$

Excluding Functions That Miss One Element in B: If a function does not map to an element in B, there are 2 elements in B left for mapping. The total number of such functions (for each specific element not mapped) is: $2^6 = 64$

Since there are 3 elements in B, the total number of such functions is: $3 \times 64 = 192$

Adding Back Functions That Miss Two Elements in B: If a function misses two elements in B, there is only 1 element left for mapping. The total number of such functions is: $1 \times 6 = 6$

Since there are 3C_2 ways to choose which two elements are missed, the total number of such functions is: 3

Using the inclusion-exclusion principle, the number of functions where all elements of B are mapped by at least one element of A is:
 $729 - 192 + 3 = 540$.

VIDEO SOLUTION

48. A

We have two equations,

$$4(x^2 + y^2 + z^2) = a \quad \text{---(1)}$$

$$4(x - y - z) = 3 + a \quad \text{---(2)}$$

Substituting the value of a from equation 1 in equation 2, we get,

$$4(x - y - z) = 3 + 4(x^2 + y^2 + z^2)$$

$$3 + 4(x^2 + y^2 + z^2) - 4(x - y - z) = 0$$

$$3 + 4x^2 + 4y^2 + 4z^2 - 4x + 4y + 4z = 0$$

It can be written as,

$$4x^2 - 4x + 1 + 4y^2 + 4y + 1 + 4z^2 + 4z + 1 = 0$$

$$(2x - 1)^2 + (2y + 1)^2 + (2z + 1)^2 = 0$$

We know that if the sum of the squares of terms is 0, then all the terms must be equal to 0

$$2x - 1 = 0$$

$$x = \frac{1}{2}$$

$$2y + 1 = 0$$

$$y = -\frac{1}{2}$$

$$2z + 1 = 0$$

$$z = -\frac{1}{2}$$

Substituting the values in equation 2, we get,

$$4\left(\frac{1}{2} - \left(-\frac{1}{2}\right) - \left(-\frac{1}{2}\right)\right) = 3 + a$$

$$4\left(\frac{3}{2}\right) = 3 + a$$

$$6 = 3 + a$$

$$a = 3$$

Therefore, the correct answer is option A.

VIDEO SOLUTION

49. 38

When given more than one equations, stating the fact that there is a common root,

We need to equate the two equations to get discernible values for x

Here, we are given three equations with the values of m, n

$$x^2 + mx + 9 = x^2 + (m + n)x + 35$$

$$mx + 9 = mx + nx + 35$$

$$nx = -26$$

Similarly, we can do it for the other equation as well,

$$x^2 + nx + 17 = x^2 + (m + n)x + 35$$

$$mx = -18$$

Substituting the value of either mx or nx in the original equations, we get

$$x^2 - 18 + 9 = 0$$

$$x^2 = 9$$

$$x = \pm 3$$

Since we are given that the root is negative, $x = -3$

$$n = -\frac{26}{-3}$$

$$m = -\frac{18}{-3}$$

$$3n = 26$$

$$2m = 12$$

$$2m + 3n = 38$$

VIDEO SOLUTION

50. A

Using the arithmetic progression formula for the n th term, where

$$x_n = a + (n - 1)d$$

Substituting the value for n and using that in the equation that is given,

$$2x_6 + 2x_9 = x_{11} + x_{13}, \text{ Then, } x_{100} \text{ equals}$$

$$\text{We get, } 2(a + 5d) + 2(a + 8d) = a + 10d + a + 12d$$

$$4a + 26d = 2a + 22d$$

$$2a = -4d$$

$$a = -2d$$

$$\text{We are given, } x_5 = -4$$

$$a + 4d = -4$$

Substituting the value for a in terms of d,

$$2d = -4$$

$$d = -2$$

$$a = 4$$

$$x_{100} = a + 99d$$

$$x_{100} = 4 - 198 = -194$$

[VIDEO SOLUTION](#)

100+ CAT Quant Questions



51.6

Let us assign the amount of work done by Renu in one hour is R

And the amount of work done by Seema in one hour is S

We are told that a certain task with different time durations for Seema and Renu

Renu=15 days with 4 hours each day, that is total work done by Renu is $15 \times 4 \times R = 60R$

Similarly for Seema, 8 days and 5 hours each day, total work done by Seema is $40S$

We know that $60R = 40S$ or $S = 1.5R$

For this task, let us assume that the amount of days that Seema decides to work is X and the hours per day that Renu decides to work is Y

We are then told that, Seema works for 2Y hours per day and Renu works for 2X days,

Work done by them will be $2XYR$ and $2XYS$

We are told that $Y=2$, Making this $4XR$ and $4XS$

$$S = 1.5R$$

Total work will be, $4XR + 6XR = 60R$

We get the value of $X = 6$

X is the number of days Seema will work, which is 6.

[VIDEO SOLUTION](#)

52. B

To find the value of $10^{100} \bmod (7)$

When 10 is divided by 7, it leaves a remainder 3, so the above equation can be written as,

$$3^{100} \bmod (7)$$

Now looking at the cyclicity of powers of 3 when divided by 7,

$$3^1 \bmod 7 = 3$$

$$3^2 \bmod 7 = 2$$

$$3^3 \bmod 7 = 6$$

$$3^4 \bmod 7 = 4$$

$$3^5 \bmod 7 = 5$$

$$3^6 \bmod 7 = 1$$

From this calculation, it is evident that the powers of 3 modulo 7 repeat every 6 steps. This forms a cycle: 3, 2, 6, 4, 5, 1

$$3^{100} = (3^6)^{16} \times (3^4)$$

$$\text{Since } 3^6 \bmod 7 = 1$$

We just need to consider $3^4 \bmod 7$ which equals 4

Hence the answer is 4.

[VIDEO SOLUTION](#)

53. C

To solve this question, we need to immediately recognise the fact that, $32768 = 8^5$

Substituting this in the above given equation,

$$\left(\frac{1}{8}\right)^k \times \left(\frac{1}{8}\right)^{5 \times \frac{1}{3}} = \left(\frac{1}{8}\right) \times \left(\frac{1}{8}\right)^{5 \times \frac{1}{k}}$$

Since the bases are equal, we can equate the powers on either side of the equation,

$$k + \frac{5}{3} = 1 + \frac{5}{k}$$

$$\frac{(3k+5)}{3} = \frac{(k+5)}{k}$$

$$3k^2 + 5k = 3k + 15$$

$$3k^2 + 2k - 15 = 0$$

Here in the given quadratic equation, the Discriminant is greater than 0, $2^2 - (4)(3)(-15) > 0$
That means both the roots are real, hence we can simply take the sum of the roots of the quadratic equation in k,
Which in a standard quadratic equation of the form $ax^2 + bx + c$ is $-\frac{b}{a}$

Here, the sum of the real values of k is $-\frac{2}{3}$

[VIDEO SOLUTION](#)

54.217

We are told that, for any natural number n_1 let a_n be the largest integer not exceeding $\sqrt{n}$

So for $n=1$, the largest integer not exceeding $\sqrt{1}$ will be 1

For $n=2$, the largest integer not exceeding $\sqrt{2}$ will be 1

For $n=3$, the largest integer not exceeding $\sqrt{3}$ will be 1

For $n=4$, the largest integer not exceeding $\sqrt{4}$ will be 2

We see a pattern here regarding the squares of the numbers,

Listing down all the perfect squares,

1, 4, 9, 16, 25, 36, 49, 64, ...

We see that the difference between 4 and 1 is 3 and there were three natural numbers in the given pattern with the value as 1,

So we can write for the rest of the numbers as well,

3 numbers will have value 1, giving a total value of 3

5 numbers will have value 2, giving a total value of 10

7 numbers will have value 3, giving a total value of 21

9 numbers will have value 4, giving a total value of 36

11 numbers will have value 5, giving a total value of 55

13 numbers will have value 6, giving a total value of 78

Now, only the values of a_{49} , a_{50} will have the value of 7, total value of 14.

Adding these values, we get the total sum as 217, which is the answer.

[VIDEO SOLUTION](#)

55.B

We are told that, the incomes of Kamal, Amal and Vimal are in the ratio 8 : 6 : 5

Lets assume them to be 80X, 60X, 50X respectively.

Money that each one of them pays towards rent,

Kamal 15% which comes to 12X

Amal 12% which comes to 7.2X

Vimal 18% which comes to 9X

Total Rental expenditure will be, 28.2X

Their incomes increase by 10%, 12% and 15% respectively,

Kamal 10% which comes to 88X

Amal 12% which comes to 67.2X

Vimal 15% which comes to 57.5X

Total income will be 212.7X

We are told the rent is the same, so it will be 28.2X

Percentage of income going towards rent in October will be, $\frac{28.2X}{212.7X} = 0.13258$

Hence, the answer is 13.26%

[VIDEO SOLUTION](#)

100+ CAT DILR Questions



56.31

When gives n distinct numbers, and asked to find the sum of the possible n distinct numbers that can be formed,

We use the formula, $(10^{n-1} + 10^{n-2} + 10^{n-3} + \dots) \times ((n-1)!) \times (\text{Sum of numbers})$

Inserting $n=4$, $(1000 + 100 + 10 + 1) (3!) (a + b + c + d)$

$(6666) (a + b + c + d)$

We are told that this value equals, $153310 + n$

Since we are told that n is a single digit natural number, the total value cannot be that much greater and 6666 should perfectly divide $153310 + n$

Upon dividing 153310 by 6666 we get the quotient as 22.99

Nearest value being 23, We take 6666×23 giving us the value 153318

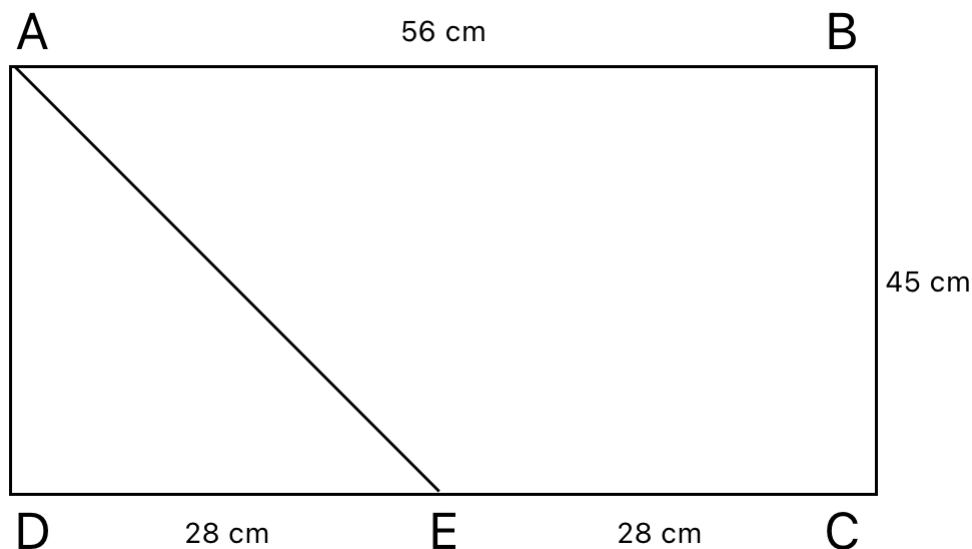
Hence the value of $n=8$ and the value of $a+b+c+d=23$

Value of $a+b+c+d+n=31$.

[VIDEO SOLUTION](#)

57.10

Drawing the figure described in the question, we get the following representation,



Since E is the midpoint of CD, the length of each half will be 28cm.

We need to find the incircle of the triangle ADE, which is a right angled triangle, we can do that with the formula,

$$\frac{(a + b - h)}{2}$$

$$h^2 = 28^2 + 45^2$$

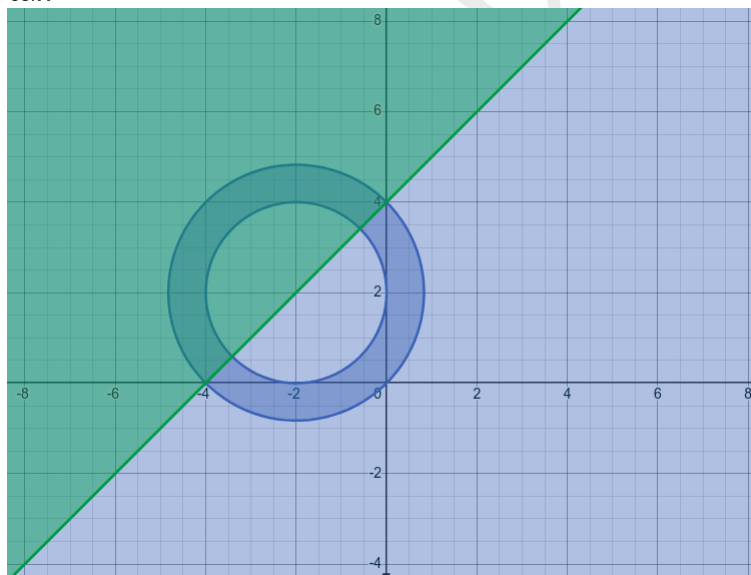
$$h^2 = 784 + 2025 = 2809$$

$$h = 53$$

$$r = \frac{(28 + 45 - 53)}{2} = 10$$

[VIDEO SOLUTION](#)

58.A



We have two inequalities,

$$y \geq x + 4$$

$$-4 \leq x^2 + y^2 + 4(x - y) \leq 0$$

The second inequality can be written as two separate inequalities,

$$-4 \leq x^2 + y^2 + 4(x - y) \text{ and } x^2 + y^2 + 4(x - y) \leq 0$$

The first inequality can be written as,

$$x^2 + y^2 + 4x - 4y + 4 \geq 0$$

$$x^2 + 4x + 4 + y^2 - 4y + 4 - 4 \geq 0$$

$$(x + 2)^2 + (y - 2)^2 \geq 4$$

The second inequality can be written as,

$$x^2 + y^2 + 4x - 4y \leq 0$$

$$x^2 + 4x + 4 + y^2 - 4y + 4 - 8 \leq 0$$

$$(x + 2)^2 + (y - 2)^2 \leq 8$$

Representing all three inequalities in the graph, we get the graph as shown above, and we must calculate the intersection of all three inequalities,

We can see that the line passes through the centre of both circles $(-2, 2)$, and the area obtained from the second inequality is the area between the two circles. So, the area of intersection of all three graphs is half of the area between both the circles as the line divides the circle in half, and we must only consider the area above the line as per the given inequality. We know that the area of the bigger circle is $\sqrt{8}$ and the area of the smaller circle is $\sqrt{4}$ from the equations of the circles as we know that equation of circle as $(x - a)^2 + (y - b)^2 = (\text{radius})^2$ where (a, b) is the centre of the circle.

The area of intersection

$$= \frac{1}{2} (\text{Area of bigger circle} - \text{Area of smaller circle})$$

$$= \frac{1}{2} (\pi (\sqrt{8})^2 - \pi (\sqrt{4})^2)$$

$$= \frac{1}{2} (8\pi - 4\pi)$$

$$= \frac{1}{2} (4\pi)$$

$$= 2\pi$$

Therefore, the correct answer is option A.

[VIDEO SOLUTION](#)

59.31

Using the logarithmic property that $\log_a b = \frac{1}{p} \log_a b$

$$4 \log_{10} x + 4 \log_{100} x + 8 \log_{1000} x = 13$$

Can be written as

$$4 \log_{10} x + 2 \log_{10} x + \frac{8}{3} \log_{10} x = 13$$

$$\frac{26}{3} \log_{10} x = 13$$

$$\log_{10} x = 1.5$$

$$x = 10^{1.5}$$

$$x = \sqrt{1000}$$

$$[\sqrt{1000}] = 31$$

Where $[\cdot]$ is Greatest Integer Function since that is what is asked in the question,

31 is the greatest integer that does not exceed x .

[VIDEO SOLUTION](#)

60.B

Let us fix the Cost Price of the product to be X , and the Selling Price of the product to be $1.4X$, since it is given that it is fixed to have a profit of 40%.

If the CP has been 40% less, making the CP $0.6X$,

And the selling price is 5 rupees less, making it $1.4X - 5$

Profit will be 50%,

$$\text{So, } 1.5 (0.6X) = 1.4X - 5$$

$$0.9X = 1.4X - 5$$

$$0.5X = 5$$

$$X = 10$$

Original selling price will be 14.

[VIDEO SOLUTION](#)

100+ CAT VARC Questions



61.B

Let us say the capacity of the glass is X , and it is completely filled with milk,

If two-thirds of its content is poured out and replaced with water, the remaining fraction of the milk will be one third.

And this is said to be done three more times, that means a total of 4 times.

So the contents of Milk initially being X ,

$$\text{And after 4 times the contents will be, } X \left(1 - \frac{2}{3}\right)^4 = \frac{X}{81}$$

Since the total contents is X , and the milk contents is $X/81$, the water contents will be $80X/81$.

Ratio of milk to water will be $\frac{X}{81} : \frac{80X}{81}$

Answer is 1 : 80

[VIDEO SOLUTION](#)

62.66

The number of apples and mangoes are in the ratio 5 : 2.

Let us write the number of apples as 5X and number of mangoes as 2X

This means oranges will be 187-7X.

After selling the remaining fruits,

Apples: 5X-75

Mangoes: 2X-26

Oranges: (187-7X)/2

Unsold Apples to Unsold oranges is 3:2

$$\frac{2(5x-75)}{187-7x} = \frac{3}{2}$$

$$20x - 300 = 561 - 21x$$

$$41x = 861$$

$$x = 21$$

Total number of unsold fruits will be,

Apples: 30

Mangoes: 16

Oranges: 20

Total is 66.

VIDEO SOLUTION

63.D

We can use the formula for when two people moving towards each other and meeting in a straight line. $t^2 = t_1 \times t_2$

Where t is time taken for them to meet each other. t_1 is the time taken by person 1 to reach the destination after meeting and t_2 is the time taken by person 2 to reach the destination after meeting

We are told they meet each other in 1 and a half hours, that is 90 minutes.

And if Sunil takes X minutes to reach A, Anil will take X+75 minutes

Since it is given that, Anil reaches B exactly 1 hour 15 minutes after Sunil reaches A

$$90^2 = x(x + 75)$$

$$8100 = x^2 + 75x$$

$$x^2 + 75x - 8100 = 0$$

$$(-75 \pm \sqrt{5625 + 32400})$$

$$\frac{(-75 + 195)^2}{2}$$

$$x = 60$$

Total time of travel of Anil is, 90 min + 60 min + 75 min

Total of 225 minutes.

Time in hours will be 3.75 hours.

$$\text{Speed is } \frac{45}{3.75} = 12 \frac{\text{km}}{\text{hr}}$$

VIDEO SOLUTION

64.15

Let us assume the four numbers to be a, b, c and d in ascending order.

Average of first two numbers is 1 more than the first number

$$\frac{(a+b)}{2} = a + 1$$

$$b - a = 2$$

$$b = a + 2$$

Average of first three numbers is 2 more than average of first two numbers

$$\frac{(a+b+c)}{3} = \frac{(a+b)}{2} + 2$$

$$2c = a + b + 12$$

Substituting the value for b

$$2c = a + a + 2 + 12$$

$$2c = 2a + 14$$

$$c = a + 7$$

Average of first four numbers is 3 more than average of first three numbers.

$$\frac{(a+b+c+d)}{4} = \frac{(a+b+c)}{3} + 3$$

$$3d = a + b + c + 36$$

Substituting the value of b and c

$$3d = a + a + 2 + a + 7 + 36$$

$$3d = 3a + 45$$

$$d = a + 15$$

d is the largest and a is the smallest and we know that d=a+15

Hence the difference between the smallest and the largest values is 15.

VIDEO SOLUTION

65.D

We are told that, 10000 is deposited in bank A for a certain number of years at a simple interest of 5% per annum.

Let us say that the number of years is x

Total value of the deposit after x years is, $10000(1 + x(0.05))$

On maturity, the total amount received is deposited in bank B for another 5 years at a simple interest of 6% per annum

Here we know the years and the interest rate,

$$10000(1 + x(0.05))(1 + 5(0.06))$$

$$10000(1 + (0.05)x)(1.3)$$

Interest received from Bank A is $(x(0.05))10000$

Interest received from Bank B is $0.3(10000(1 + x(0.05)))$

This ratio is given to be 10:13.

$$\frac{x(0.05)}{0.3(1 + x(0.05))} = \frac{10}{13}$$

$$0.65x = 3 + 0.15x$$

$$0.5x = 3$$

$$x = 6$$

Hence the number of years the money was invested in Bank A is 6 years.

VIDEO SOLUTION

CAT Expected Questions PDF



66.C

Let us say the quantity of grains is X

For the first customer he sells $\frac{X}{2} + 3$

Remaining is $\frac{X}{2} - 3$

Second customer he sells: $\frac{X}{4} - \frac{3}{2} + 3 = \frac{X}{4} + \frac{3}{2}$

Remaining will be $\frac{X}{4} - \frac{9}{2}$

Third customer he sells: $\frac{X}{8} - \frac{9}{4} + 3 = \frac{X}{8} + \frac{3}{4}$

Remaining will be $\frac{X}{8} - \frac{21}{4}$

Now, this is said to be 0,

$$\frac{X}{8} - \frac{21}{4} = 0$$

$$X = 42$$

VIDEO SOLUTION

67.A

$(a + b\sqrt{n})$ is the positive square root of $(29 - 12\sqrt{5})$

$$\text{So } 29 - 12\sqrt{5} = (a + b\sqrt{n})^2$$

$$29 - 12\sqrt{5} = a^2 + b^2n + 2ab\sqrt{n}$$

$$a^2 + b^2n = 29 \text{ and}$$

$$ab\sqrt{n} = -6\sqrt{5}$$

$$a^2b^2n = 180$$

$$b^2n = \frac{180}{a^2}$$

Substituting this in the above equation,

$$a^2 + \frac{180}{a^2} = 29$$

$$a^4 - 29a^2 + 180 = 0$$

$$a^2 = \frac{(29 \pm \sqrt{29^2 - 4(180)})}{2}$$

$$a^2 = \frac{(29 + \sqrt{841 - 720})}{2}$$

$$a^2 = 9 \text{ or } 20$$

That means, one of a^2 or b^2n is 9 and 20.

We also have, $ab\sqrt{n} = -6\sqrt{5}$ that means one of a or b should be negative

And also the fact that this is a positive square root,

And we need to maximise the value of a , b and n .

We can have $a=3$, $b=1$ and $n=20$.

This satisfies all the above equations, and the value of $a+b+n=18$.

VIDEO SOLUTION

68.C

Given that, The surface area of a closed rectangular box, which is inscribed in a sphere, is 846 sq cm

$$\text{So, } 2(lb + bh + hl) = 846.$$

$$\text{And } 4(l + b + h) = 144$$

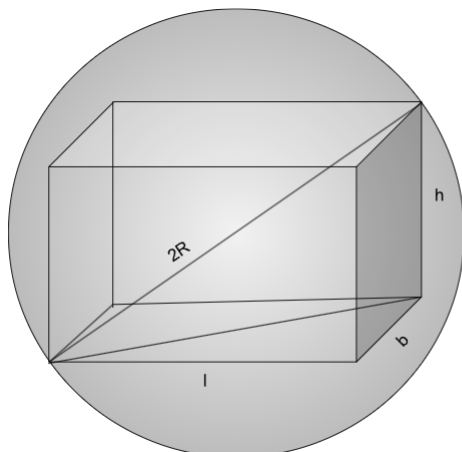
$$(l + b + h) = 36$$

$$(l + b + h)^2 = l^2 + b^2 + h^2 + 2(lb + bh + hl)$$

$$1296 = (l^2 + b^2 + h^2) + 846$$

$$450 = l^2 + b^2 + h^2$$

We are told that this cuboid is inscribed in a sphere, the body diagonal of the cuboid equals the diameter of the sphere, this can be visualised as:



This is nothing but, $\sqrt{l^2 + b^2 + h^2} = 2R$

$$l^2 + b^2 + h^2 = 4R^2$$

$$450 = 4R^2$$

$$R^2 = \frac{225}{2}$$

$$R = \frac{15}{\sqrt{2}}$$

$$\text{Volume of sphere will be } \frac{4}{3} \times \pi \times \left(\frac{15}{\sqrt{2}}\right)^3$$

$$\frac{4}{3} \pi \left(\frac{3375}{2\sqrt{2}}\right)$$

$$\pi \times 1125\sqrt{2}$$

[VIDEO SOLUTION](#)